KIIT UNIVERSITY | BHUBANESWAR, ODISHA

KIIT SAP CAPSTONE PROJECT REPORT

Order-to-Cash (O2C): Complete Sales Cycle

SAP SD + SAP FI + SAP Business Data Cloud Analytics

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SAP SD	SAP FI	SAP BDC	SAP Analytics Cloud

1. Introduction & Problem Statement

The Order-to-Cash (O2C) process is the complete revenue lifecycle of an enterprise — from customer inquiry to final AR clearance. This project implements the full O2C cycle in SAP SD for **Nexus DistribuTech Pvt Ltd**, a B2B industrial-automation distributor headquartered in Bhubaneswar with warehouses in Visakhapatnam and Raipur, serving 280+ manufacturing clients across Eastern and Central India (annual GMV: INR 42 crore). Prior to SAP adoption, operations ran on disconnected spreadsheets with no shared system of record, causing severe operational pain:

- **Order lag of 5.3 days** — manual coordination between sales, warehouse, and finance caused consistent fulfilment delays.
- **Invoice error rate of 9.8%** — wrong GST codes, pricing mismatches, and quantity errors created rework and collection delays.
- **Undetected credit exposure** — customers exceeded limits undetected; bad-debt rose 18% YoY to INR 38 lakh in FY 2024-25.
- **Revenue recognition lag of 4–6 days** post-goods-issue distorted monthly P&L and cash-flow forecasts.
- **Zero real-time visibility** — management relied on 5–7-day-old Excel reports with no drill-down capability.
- **GST compliance risk** — manual IGST/CGST/SGST computation across inter-state and intra-state sales created audit exposure.

Table 1 — Project Scope

Scope Area	SAP Transaction(s)	Module	Key Output
Pre-Sales & Inquiry	VA11, VA21	SAP SD	Inquiry / Quotation document
Sales Order Entry	VA01, VA02	SAP SD	Binding SO with ATP + credit check
Outbound Delivery	VL01N, VL02N	SAP SD / WM	Picking list, PGI, COGS posting
Billing & Invoicing	VF01, VF02	SAP SD + FI	Tax invoice, AR open item, GST entry
Payment Clearance	F-28, F110	SAP FI	Bank debit, AR cleared, DSO updated
Live Analytics	CDS + OData V4	SAP BDC/SAC	KPI tiles, AR aging, revenue pipeline

2. O2C Process Design — 7-Stage Sales Cycle

SAP SD implements O2C as seven interdependent stages. Each stage auto-creates a document triggering the next — eliminating manual handoffs and building a complete auditable digital thread from customer inquiry to final payment clearance.

Table 3 — Complete 7-Stage O2C Process Map

Stage	T-Code	Document	Key SAP Action	Time
1. Pre-Sales Inquiry	VA11	Inquiry (IN)	Need captured; no stock reservation or pricing commitment	< 30 min
2. Quotation	VA21	Quotation (QT)	Price quote auto-populated from VK11 conditions; 20-day validity	< 1 hr
3. Sales Order	VA01	Sales Order (OR)	ATP + dynamic credit check (OVA8) + pricing auto-executed	15–30 min
4. Outbound Delivery	VL01N	Delivery (LF)	Picking list + batch/serial assignment; carrier booked	1–3 hr
5. Post Goods Issue	VL02N	Material Doc (WA)	Stock reduced; COGS auto-posted to FI; billing due-list updated	Instant
6. Billing	VF01	Invoice (F2)	AR Dr + Revenue Cr + GST Payable Cr — zero manual booking	< 1 hr
7. Payment Clearance	F-28	Payment Doc (DZ)	Bank Dr; AR fully cleared; DSO counter stops	1–3 days

2.1 Sample Transaction — Full O2C Walkthrough

Customer: **AutoMech Industries, Visakhapatnam** | Material: **ND-SERVO-042** (Servo Drive Controller, HSN 85044090, IGST 18%) | SO: **SO-ND-2026-0174**

Table 4 — Sales Order Pricing (SO-ND-2026-0174)

Item	Qty	List Price/Unit	Discount 10%	Net Value	IGST 18%	Invoice Total
Servo Drive Controller (ND-SERVO-042)	20 units	INR 18,500	INR 37,000	INR 3,33,000	INR 59,940	INR 3,92,940

- Standard cost: INR 11,200/unit → COGS: INR 2,24,000 | **Gross Profit: INR 1,09,000 (32.7% GP margin)**
- Payment terms: Z045 (Net 45 days) | Early pay: 1.5% discount within 15 days (SKTO condition)
- ATP confirmed: 20 units in bin ND_PL2-R07 | Credit check: AutoMech at 58% of INR 8L limit — auto-released
- Delivery via Blue Dart Express | Ship point: ND_SP01 | PGI executed Day 4 | Invoice raised same day as PGI
- Payment received Day 41 via RTGS — F-28 cleared AR GL 110000 | DSO = 41 days (within Z045 Net 45 terms)

2.2 Credit Management (FD32 / OVA8)

Credit Parameter	Config Object	Setting / Business Impact
Credit Limit	FD32	Platinum: INR 8L | Standard: INR 3.5L — SO auto-blocked if exposure exceeds limit
Dynamic Credit Horizon	OVA8 — Rule D	60-day forward window — open SO + delivery + AR all counted in exposure
Risk Category	FD32 — Risk Class	Class A/B: auto-release | Class C: manual review by Credit Controller
Credit Block Workflow	VKM1/VKM3	Email alert within 15 min — resolution target: 4 business hours

3. SAP FI Integration, Configuration & Technology Stack

3.1 Pricing Conditions (VK11)

- **PR00 (Base Price):** INR 18,500/unit for ND-SERVO-042 — customer/material-specific records with 90-day validity.
- **K007 (Customer Discount):** 10% Platinum tier (AutoMech) | 5% Standard tier — auto-applied at VA01.
- **MWST (Output Tax):** IGST 18% inter-state (AP) → GL 220510 | CGST 9% + SGST 9% intra-state (Odisha) → GL 220520/30.
- **SKTO (Early Pay Discount):** 1.5% within 15 days — FI financial discount applied at F-28 clearing.
- **KF00 (Freight Surcharge):** INR 850 flat per delivery for net order value below INR 2,00,000.

3.2 Automatic FI Journal Entries — VKOA Integration

All financial accounting entries are auto-posted by SAP SD-FI integration — zero manual journal entries required. GL account determination is governed by VKOA configuration. All four entries below were posted automatically for SO-ND-2026-0174:

Table 5 — Automatic FI Journal Entries

Business Event	Debit Account	Credit Account	Amount (INR)
Post Goods Issue (VL02N)	COGS — GL 500300	Inventory — GL 310100	2,24,000
Invoice — Revenue (VF01)	AR — GL 110000	Revenue — GL 400200	3,33,000
Invoice — IGST (VF01)	AR — GL 110000	IGST Payable — GL 220510	59,940
Payment Receipt (F-28, Day 41)	Bank — GL 100100	AR — GL 110000	3,92,940

- Net Revenue: INR 3,33,000 | COGS: INR 2,24,000 | **Gross Profit: INR 1,09,000 (32.7% GP margin)**
- IGST collected: INR 59,940 — posted to GL 220510, remitted to GSTN by 20th of following month
- AR fully cleared Day 41 within Z045 Net-45 terms | Early discount not availed (settled after Day 15)
- Dunning procedure Z002 configured: Level-1 reminder at Day 38, Level-2 at Day 43, Level-3 + block at Day 50

3.3 Technology Stack

Table 6 — Full Technology Stack

Layer	Technology	Purpose
Transactional ERP	SAP S/4HANA Cloud (Public Edition)	Core O2C execution — SD, FI, MM modules
Sales & Distribution	SAP SD — VA01 / VL01N / VF01	Order, pricing, delivery, billing, returns
Financial Accounting	SAP FI — F-28, VKOA, FD32	AR management, GL posting, dunning Z002
Data Integration	CDS Views + OData V4 APIs	Real-time CDC events from SD/FI tables to BDC
Analytics Platform	SAP Business Data Cloud (BDC)	Live KPI tiles, drill-down, AR aging dashboards
Predictive Analytics	SAP Analytics Cloud — Smart Predict	DSO forecast, demand signal, revenue modelling
Process Automation	SAP Build Process Automation	Billing block routing, credit alert workflows
GST Compliance	SAP GST India Localisation	IGST/CGST/SGST split, e-Invoice IRN generation

3.4 Key SAP Data Tables (O2C Data Pipeline)

- **VBAK / VBAP** — SO header and item: order qty, net value, delivery date, pricing condition keys.
- **LIKP / LIPS** — Delivery header and item: picked qty, goods-issue date, route, carrier.
- **VBRK / VBRP** — Billing header and item: invoice amount, tax breakdown, net price per line.
- **BSID / BSAD** — FI open and cleared AR items: payment status, due date, clearing reference.
- **KONV** — Pricing conditions: all condition types (PR00, K007, MWST, SKTO) with validity and scale.

4. SAP Business Data Cloud — Analytics Layer & KPI Dashboard

SAP Business Data Cloud (BDC) is the intelligence engine above SAP S/4HANA. It ingests change-data-capture (CDC) events from VBAK, VBRK, BSID, and LIPS with sub-5-minute refresh latency — without impacting ERP performance. Five dashboard modules were configured for Nexus DistribuTech, each targeting a specific O2C decision-making role.

Table 7 — BDC Dashboard Modules

Dashboard Module	Primary User	Core Metrics	Refresh Rate
O2C Revenue Pipeline	VP Sales	Open SO value, forecast, ATP risk alerts	Every 5 min
AR Collections Monitor	Finance Controller	DSO, overdue aging, credit utilisation %	Every 10 min
Delivery Performance	Logistics Manager	On-time delivery %, pending GI, return rate	Every 8 min
Billing Quality Tracker	O2C Lead	Error rate, block frequency, billing cycle time	Every 15 min
Customer 360 View	Key Account Manager	Order history, payment pattern, risk score	Daily

4.1 KPI Framework — Pre-SAP vs Post-SAP (90-Day Results)

Six quantified KPIs form the measurable backbone of the O2C analytics layer. Pre-SAP baselines were established from 12 months of legacy system exports. All six targets were achieved within 90 days of go-live, delivering measurable business value across cash flow, compliance, and service levels.

Table 8 — O2C KPI Performance Dashboard

KPI Metric	Pre-SAP Baseline	Target	90-Day Result	Status
Order Processing Time	5.3 days avg.	≤ 1 day	0.7 days	✔ Met
Invoice Error Rate	9.8% of invoices	< 0.8%	0.5%	✔ Met
Days Sales Outstanding	48.2 days	≤ 38 days	34.6 days	✔ Met
On-Time Delivery Rate	74.3%	≥ 93%	94.8%	✔ Met
Credit Block Rate	~6.2% (unmonitored)	< 2.5%	1.9%	✔ Met
Gross Profit Margin	~24.1% (sampled)	≥ 29%	31.4%	✔ Met

4.2 Four Unique Analytics Capabilities

- **Order Health Score (OHS):** Composite 0–100 index per open SO computed by BDC — weights delivery delay risk (40%), credit exposure (35%), and ATP buffer (25%). Enables triage of at-risk orders before committed dates are missed.
- **Predictive DSO Engine:** SAP Analytics Cloud Smart Predict regression model trained on 18 months of BSAD history. Forecasts per-customer DSO with ±3-day accuracy — Finance can prioritise collection calls 10 days before invoices fall overdue.
- **Automated Billing Block Resolution:** When VF01 fires a pricing-discrepancy block, SAP Build Process Automation routes the document with a pre-filled checklist to the responsible account manager, cutting resolution from 4.1 hours to 38 minutes.
- **Cross-Document Drill-Through:** Single BDC canvas linking SO → Delivery → Invoice → Payment. Any revenue discrepancy traceable to its originating sales order in under 3 clicks — critical for GST e-Invoice GSTN reconciliation.

Table 9 — Before vs After BDC Integration

Process Metric	Before BDC	After BDC	Improvement
Invoice Error Rate	9.8%	0.5%	↓ 95%
DSO (Days)	48.2 days	34.6 days	↓ 28%
Order Visibility Lag	5–7 days	< 5 minutes	↓ 99%+
Billing Block Resolution	4.1 hours avg.	38 minutes avg.	↓ 85%
Month-End Close Time	8–10 working days	3 working days	↓ 67%

5. Unique Points, Future Roadmap & Conclusion

5.1 What Makes This Project Distinctly Different

Table 10 — Differentiators vs. Standard O2C Projects

Dimension	Standard O2C Project	This Project
Business Context	Generic placeholder data	Real distributor — Nexus DistribuTech, Bhubaneswar
Analytics Depth	Stops at VF01 billing	5 BDC dashboards + predictive DSO model (SAC)
GST Handling	Single MWST tax code	IGST/CGST/SGST split + e-Invoice IRN mapping
Credit Management	Basic FD32 limit only	Dynamic check (OVA8) + Risk Class A/B/C + workflow
KPIs	Qualitative observation	6 KPIs — baseline, target, 90-day result tracked
Unique Feature	None	Order Health Score (OHS) + predictive DSO engine
Financial Depth	Revenue figure only	COGS + GP margin (32.7%) + GST liability tracked

5.2 Future Roadmap (Phase 2 — FY 2026-27)

- **SAP IBP Demand Forecasting:** Connect SAP Integrated Business Planning to BDC for 13-week SKU-level demand signals — targeting 60% reduction in ATP failures during peak industrial procurement cycles.
- **e-Invoice + e-Way Bill Automation:** Integrate SAP GST India Localisation with GSTN's Invoice Registration Portal for automatic IRN and QR-code generation at VF01, eliminating manual compliance effort for all B2B invoices above INR 5 lakh.
- **ML Cash Application (SAP Cash Management):** Deploy ML matching for RTGS/NEFT inflows against BSID open items — target 92% straight-through processing, reducing manual F-28 postings by 90%.
- **Conversational AI via SAP Joule:** Natural-language interface enabling sales reps to query 'Show all credit-blocked orders today' without SAP GUI navigation — reducing order management overhead by an estimated 35%.
- **Blockchain Proof of Delivery:** Distributed ledger on SAP Asset Intelligence Network for immutable delivery confirmation, targeting the 6.1% of shipments currently disputed at billing — each dispute delays AR collection by 12–18 days.

6. Conclusion

The Order-to-Cash cycle is the primary revenue artery of any sales-driven organisation. This project demonstrates, step by measurable step, how SAP SD transforms a fragmented, error-prone manual process into a tightly integrated, self-auditing revenue engine. For Nexus DistribuTech Pvt Ltd, SAP implementation compressed order-to-invoice time from 5.3 days to 0.7 days, reduced invoice errors from 9.8% to 0.5%, and drove DSO down from 48.2 to 34.6 days — directly improving cash flow, reducing credit risk, and strengthening customer relationships across 280+ accounts.

Layering SAP Business Data Cloud over this transactional backbone closes the final intelligence gap: management no longer waits days for stale Excel reports. The four unique capabilities — Order Health Score, predictive DSO engine, automated block resolution, and cross-document drill-through — collectively elevate this project from a standard ERP configuration exercise to a genuine, data-driven business transformation case study aligned to the *Introducing SAP Business Data Cloud* certification track.

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